The Case for Demand Reduction

Last May I retired from the Federal Energy Regulatory Commission after 23 years. Part of my job was to take care of public inquiries. In my last two years of employment I talked to numerous citizens about their growing despair over high electric bills. I particularly remember a couple of calls where women, in tears, told me that they would have to choose between paying their exploding electric bill or buying food.

I tried to explain to them the larger market forces that were driving prices up. But nobody was buying it. To the individual householder, nothing had changed. Their electric usage was the same as last year. They were all convinced that rising prices were the result of greedy and corrupt utility executives. The simple truth is that people love the free market as long as prices are going down.

Some years ago I listened to a speech given by Dr. Steven Chu, a Nobel laureate who served as Secretary of Energy. Dr. Chu commented that the U.S. could forego building multiple generating plants if people just painted their roofs white.

When I retired I had the opportunity to test Dr. Chu's assumption. The shingle roof of my house was reaching the end of its 20-year life. Instead of putting on a new roof, I opted to paint it with special products that are designed to extend the life of the shingles. But instead of a clear coating, which comprise the majority of commercial products, I chose a UV-protective product, turning my medium-gray shingles to an off white. Basically, a light roof just absorbs less energy than a dark roof. Everyone knows that's how it works.

But I also became aware of a new concept called a "cool roof." If a paint is impregnated with microscopic spheres it will literally reflect UV rays back into space. This painted surface will actually be cooler than the ambient air temperature surrounding it. After looking into various options I discovered a company that produces microscopic glass beads that have the consistency of fine sand. This company is called Guerra Paint and Pigment in New Jersey. (I'm sure there are other suppliers as well).

The glass beads can be added to any paint to render it light reflective. The primary consumer of this product is the Department of Motor Vehicles, who add it to the paint they use to mark stripes on streets and freeways. Even in areas where there are no street lights, if this paint is used the driver can easily see where the road is from the light that is reflected off the road from their head lights. A pound of this product cost just $42.

I painted the roof of my three-bedroom home two coats of my "mist gray" paint, and with a couple of gallons of leftover paint I added the glass beads and used that as an extra coat on the south side of my roof.

The results were immediate and dramatic. My house suddenly seemed calmer. I realized that was because the furnace fan wasn't constantly humming. My upstairs bedrooms were now much cooler. The attic fan, which turns on when that space reaches 95 degrees and would keep me awake all night, no longer turned on.

I don’t have kilowatt statistics, but anecdotally I can confirm Dr. Chu’s assumptions.

But how does this translate to the larger issue at hand? Clearly, converting individual homes would take time and considerable effort. The average roof lasts 20 years, so conversion would be gradual. Any program focused on houses would also have to contend with “aesthetic” issues about individual tastes about white roofs. Some homes have slate roofs or ceramic tiles that are tied to historic preservation. There are all kinds of issues.

The low-hanging fruit for demand reduction lies with flat-roofed commercial buildings, most of which have a roofing system of black rubber. These systems do a fine job preventing moisture infiltration to the building, but it also means that the structure itself is a heat sink. Then the building occupants compensate for the heat that the sun creates by cranking up the air conditioning.

Fortunately, there is a low-cost solution to this problem.  There’s no need to build new roofs on these buildings. I found several commercially-available products that can convert these black rubber roofs to a UV reflective system through the application of an elasticised paint. The process is straightforward. Clean off the roof–a power wash will do the trick. Then paint one layer of base paint, followed by a top coat of bright white. The final coat needs 24 hours to dry. You could even add glass beads to aim for a “cool” roof effect.

Buying the paint supplies for my roof cost about $1,500 plus $42 for glass beads. When I priced out the elasticised paint I figured that the cost of elasticised paint for a commercial building would probably be between $3,000 - $5,000. These costs would fall when operated in a program that exploits economies of scale in purchasing supplies.

The cost of demand reduction would be pennies on the dollar compared to the cost of new generation. And there are literally millions of flat-topped buildings in the U.S., office buildings, municipal buildings housing federal, state and local bureaucracies, libraries, schools. And most importantly, high-rise apartment houses.

In Arlington, Virginia where I live, over half of our population live in multi-family units, a huge number of which are flat-topped high-rises. The owners and managers of those buildings are less likely to pursue a UV-reflective roof because the electric usage is individually-metered in the separate apartments. But this is where most of the working-class and non-home owners live, the very people who are in desperate need of respite from high electric costs.

In the larger market, significant demand reduction will create downward pressure on electric wholesale prices, which will benefit everyone, including the large users like the data centers. In the end, everyone wins.

We should not discount the importance of demand reduction. For almost 20 years the U.S. had flat demand, which kept electricity prices affordable across most of the country. Not enough attention or recognition was afforded to the U.S. Department of Energy's "Energy Star" program, which worked to improve the efficiency of electric appliances.

During this time American consumers were quietly swapping out their old, inefficient refrigerators, stoves, clothes dryers, washing machines, dish washers and other items with models that saved energy. This quiet evolution meant that, even though the population grew, energy consumption remained steady.

We're in a new era of dramatic increases in demand. We won't be able to build our way out of this mess. Small nuclear units are still years away, gas generating equipment is not immediately available for new construction. Demand reduction must be considered at least as a partial solution.

My proposal is simple. The data centers and other large users should be required, as a condition of interconnection, to secure a commensurate amount of demand reduction to the amount of load they plan to put onto the grid. This would be consistent with the ratemaking principle of "cost causation."

I would suggest that they start this effort in the very communities that have been hardest hit by their presence–in the communities that already have clusters of data centers. One way to achieve this would be to offer local governments, commercial property owners, and individual households, the free service of converting their roofs to "cool roof" technology, or at least to UV-reflective use. I believe this program would be a welcome relief.

To those who argue that my solution only addresses summertime use, I am confident that our industry is sufficiently innovative to find additional opportunities for demand reduction. We just need to start thinking about this problem in a different way.

I offer my best wishes to the Commission in their deliberations on this very important topic.

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